

Financial Tweet Sentiment Classification

Multiclass Sentiment Analysis of Financial Social Media Text

Text Mining — MSc in Data Science and Advanced Analytics

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Abstract

This report presents a multiclass sentiment classification system for financial tweets, distinguishing three market stances: Bearish (0), Bullish (1), and Neutral (2). The dataset comprises 9,543 labelled training tweets with pronounced class imbalance (Neutral 64.7%, Bullish 20.1%, Bearish 15.1%; imbalance ratio 4.28:1). We systematically evaluate six text preprocessing techniques, eleven feature representations — spanning sparse TF-IDF, Word2Vec, GloVe-Twitter, FinBERT, SBERT and Twitter-RoBERTa embeddings — and twelve model families, all under an identical 5-fold stratified cross-validation protocol. Frozen-embedding classifiers tuned with Optuna plateau around macro-F1 0.80. We then conduct a **transformer backbone search** (model selection via the Hugging Face Hub), end-to-end fine-tuning four encoders; the best, a Twitter-domain **RoBERTa-large** (TimeLMs; Loureiro et al., 2022), reaches an out-of-fold macro-F1 of **0.8859** — a **+0.0838** gain over the 0.8021 baseline and our **submitted single-model solution**. Notably, this single model also surpasses a six-learner stacking ensemble (extra work, 0.8483), so the submission is simultaneously our best result overall. We additionally show that a fine-tuned **GPT-2 decoder** classifies at macro-F1 0.7724 (whereas few-shot prompting collapses to the majority class), and that an offline **agentic workflow** adaptively routes tweets between VADER, LightGBM/SBERT and FinBERT with conversational explanations. An ablation confirms that preserving raw financial text outperforms every preprocessing pipeline.

Keywords: Financial sentiment analysis, tweet classification, Twitter-RoBERTa fine-tuning, model selection, decoder classification, stacking, LangChain agent.

1 Introduction

Social media platforms — particularly Twitter/X — have become a primary channel through which retail and institutional investors express and discover market sentiment. Automated classification of financial tweets into Bearish, Bullish and Neutral stances enables real-time signal generation for algorithmic trading, risk monitoring and investor-sentiment tracking (Bollen et al., 2011; Antweiler & Frank, 2004).

This work addresses multiclass financial sentiment classification under significant class imbalance (ratio 4.28:1). We explore the full NLP pipeline — from raw text through feature engineering to model selection and hyperparameter tuning — following the Text Mining curriculum at NOVA IMS. All models are trained *only* on the provided training corpus. The remainder of the report is organised as follows: Section 2 explores the dataset; Section 3 analyses preprocessing; Section 4 describes feature engineering; Section 5 presents the classification models, including the backbone search and the fine-tuned encoder/decoder; Section 6 reports results and error analysis; Section 7 details

the agentic workflow (Extra Challenge 2); Section 8 concludes.

Course alignment. Every stage follows the Text Mining curriculum: the NLP pipeline and text preprocessing (Lecture 1, Lab 1); n-grams, TF-IDF and distance metrics (Lecture 2-Extra); word2vec word embeddings and their semantic geometry (Lecture 2, Lab 2); attention and Transformer *encoders* (Lecture 4, Lab 4); and *decoder* LLMs, encoder-based classification and *Agentic AI* (Lecture 6, Lab 5; Alammar & Grootendorst, 2024).

2 Data Exploration

The dataset consists of 9,543 labelled training tweets and 2,388 unlabelled test tweets from financial Twitter discourse. Each tweet references one or more publicly-listed securities and expresses a market stance.

2.1 Class Distribution

Figure 1 illustrates the severe class imbalance. Neutral tweets dominate (64.7%), consistent with most financial discourse being non-directional. Bearish tweets (15.1%) are less frequent than Bullish (20.1%), reflecting a structural optimism bias in financial social media (Antweiler & Frank, 2004). To address imbalance, all models use `class_weight='balanced'` (or an inverse-frequency-weighted loss) and evaluation relies on **macro-F1** as the primary metric, which weights all three classes equally regardless of support.

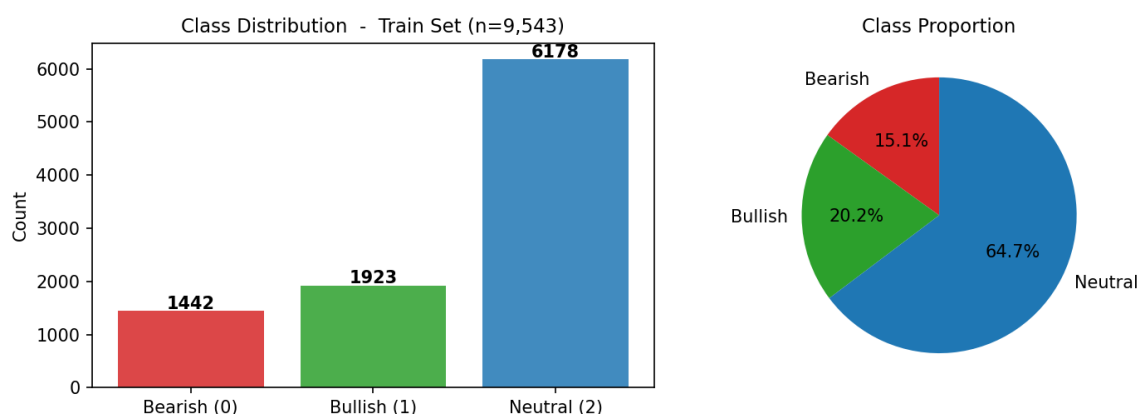


Figure 1: Class distribution of the training set: absolute counts (left) and proportions (right).

2.2 Linguistic Characteristics

Tweets have a mean length of 12.2 words (maximum 32), consistent with Twitter's character limit. Key Twitter-specific features include cashtags (\$AAPL, \$TSLA), hashtags (#earnings, #bearish), mentions (@analyst) and URLs. Bullish tweets contain more cashtags on average (0.34 vs. 0.22 for Bearish), suggesting that positive sentiment correlates with specific stock discussion. The most discriminative terms per class are: **Bearish** — 'down', 'miss', 'cut', 'bearish'; **Bullish** — 'beats', 'strong', 'upgraded', 'buy'; **Neutral** — 'flat', 'unchanged', 'stable'. Per-class word clouds are included in the accompanying notebook.

2.3 EDA-to-Modelling Decisions

The exploratory analysis is not only descriptive; it directly determines the modelling choices.

Finding	Evidence	Modelling consequence
Severe class imbalance	Neutral 64.7%, Bullish 20.1%, Bearish 15.1%	Stratified folds, class-weighted losses and macro-F1 as the primary metric.
Short, information-dense tweets	Mean 12.2 words, maximum 32 words	Transformer max length 96 covers the corpus with headroom and avoids unnecessary padding.
Financial tokens carry sentiment	Cashtags, analyst actions, earnings and target-price terms	Raw text is preserved for the final transformer; aggressive cleaning is tested but not assumed.
URL/mention boilerplate is common	URLs appear in roughly half of the tweets	Regex cleaning is evaluated for sparse baselines, while tweet-specialised tokenizers handle raw text.

Table 1: How EDA findings are translated into concrete modelling choices.

2.4 Corpus Split and Validation Protocol

To satisfy the corpus-split requirement without wasting labelled data, every experiment uses **5-fold stratified cross-validation** on the 9,543 labelled tweets (`StratifiedKFold`, `shuffle=True`, `random_state=42`). Stratification keeps the Bearish/Bullish/Neutral proportions stable in each validation fold, which is essential under the 4.28:1 imbalance. The final notebook keeps the same protocol: each fold trains on 80% of the labelled corpus, validates on the remaining 20%, stores out-of-fold predictions for honest macro-F1, and averages the five test-set probability vectors to produce `pred_33.csv`.

3 Data Preprocessing

We implement six preprocessing techniques following the NLTK-based approach from course Lab 1 and the tweet-tokenisation practice from the extra TF-IDF lab. Crucially, preprocessing is treated as an *empirical question*: every variant is measured under the same 5-fold protocol before deciding what enters the final pipeline.

3.1 Preprocessing Pipeline

- **T1 — Twitter noise cleaning.** Regex removal of URLs and mentions ($\rightarrow$ USER); hashtag-symbol stripping; cashtags mapped to TICKER_X.
- **T2 — Unicode normalisation.** NFKD decomposition then ASCII encoding.
- **T3 — Stop-word removal.** NLTK English list, negations preserved (`{not, no, never, neither, nor, none}`) — critical for sentiment (Manning et al., 2008).
- **T4 — WordNet lemmatisation.** POS-aware: ‘beaten’ $\rightarrow$ ‘beat’, ‘falling’ $\rightarrow$ ‘fall’.
- **T5 — Snowball stemming.** More aggressive: ‘earnings’ $\rightarrow$ ‘earn’.
- **T6 — TweetTokenizer.** `reduce_len=True`, `strip_handles=True`.

Example transformations are shown in the notebook: URLs and mentions become generic tokens, cashtags retain the ticker identity, negations survive stop-word removal, and lemmatisation/stemming reduce inflected forms. This demonstrates compliance with the preprocessing requirement while also exposing the main trade-off: cleaning removes noise, but may also remove finance-specific signal.

3.2 Ablation Study

Each configuration is evaluated with Logistic Regression ($C = 1.0$) and TF-IDF (10,000 vocabulary) under 5-fold stratified CV.

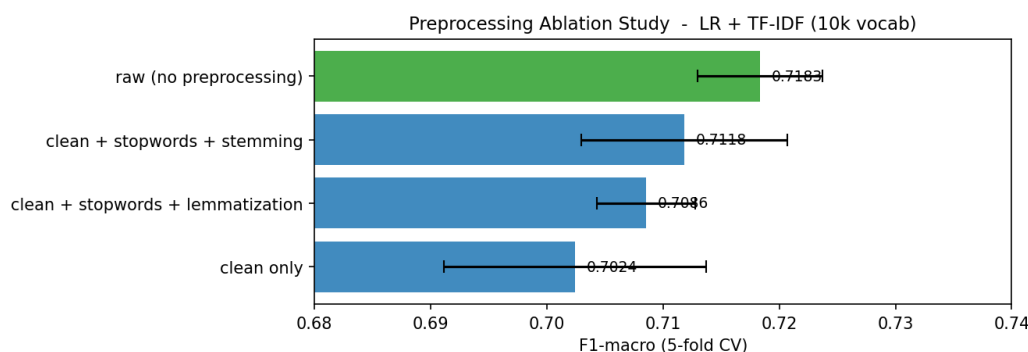


Figure 2: Preprocessing ablation: macro-F1 by configuration (green = best). Raw text wins.

Preprocessing configuration	Macro-F1	Std
raw (no preprocessing)	0.7183	0.0053
clean + stop-words + stemming	0.7118	0.0089
clean + stop-words + lemmatisation	0.7086	0.0042
clean only	0.7024	0.0113

Table 2: Preprocessing ablation (5-fold CV, LR + TF-IDF). Raw text (shaded) is best.

Key finding: raw text achieves the highest macro-F1 (0.7183). This is consistent with prior financial NLP work (Go et al., 2009): ticker symbols (\$AAPL), analyst-firm names and domain vocabulary (‘downgrade’, ‘EPS beat’, ‘guidance raise’) carry strong discriminative signal that preprocessing destroys. Therefore, the project does *not* skip preprocessing; it implements, documents and ablates it, then justifies the final raw-text choice with evidence. Sparse baselines still use the cleaned variants for comparison, while the submitted Twitter–RoBERTa-large relies on its native tweet tokenizer and receives raw financial text.

4 Feature Engineering

Eleven feature representations are computed, covering the three families required by the brief: bag-of-words, Word2Vec and Transformer Encoder.

4.1 Sparse Representations (BoW)

Four sparse sets on raw text: **BoW** (binary CountVectorizer, max_features=20k, min_df=2); **TF-IDF 1g** (max_features=20k, max_df=0.80, min_df=2); **TF-IDF 2g** (uni+bigram, max_features=50k); and **TF-IDF char** (char 3–5-gram, max_features=30k).

4.2 Word Embeddings (Word2Vec / GloVe)

Word2Vec Skip-gram and **CBOW** (200d) trained on the corpus (Mikolov et al., 2013), pooled by TF-IDF-weighted mean; and **GloVe-Twitter-100** pre-trained on 2B tweets (Pennington et al., 2014).

4.3 Transformer Embeddings (Encoder)

Three transformer representations (CLS / mean-pooling): **FinBERT** (Araci, 2019); **SBERT all-mpnet-base-v2** (Reimers & Gurevych, 2019; +0.50 extra); and **Twitter–RoBERTa** (Barbieri et al., 2020; +0.50 extra). The latter two count as additional Transformer Encoder methods for the extra-work criterion, and the Twitter–RoBERTa family is carried forward to end-to-end fine-tuning in Sec-

tion 5.2.

4.4 Financial Hand-crafted Features

A 14-dimensional vector combines VADER scores (Hutto & Gilbert, 2014) with surface statistics (cashtag/hashtag/mention/URL counts, length, capitalisation, punctuation) and a Flesch–Kincaid readability score.

5 Classification Models

All models are evaluated via 5-fold stratified cross-validation (StratifiedKFold, random_state=42) with macro-F1 as the primary metric; the identical split across every experiment makes the out-of-fold (OOF) scores directly comparable.

5.1 Traditional ML Models

- **Naïve Bayes.** MultinomialNB ($\alpha = 1.0$) and ComplementNB ($\alpha = 0.5$) on BoW / TF-IDF; ComplementNB targets imbalance (Rennie et al., 2003).
- **Logistic Regression.** L_2 (SAGA), $C \in \{0.1, 1.0, 10.0\}$, all feature sets.
- **Linear SVM.** CalibratedClassifierCV(LinearSVC), $C \in \{0.1, 1.0\}$.
- **KNN.** $k \in \{5, 15\}$, cosine distance, dense embeddings.
- **MLP.** (256, 128) and (512, 256, 128), ReLU, early stopping.
- **Random Forest / XGBoost / LightGBM.** 100 and 300 estimators; LightGBM is the strongest traditional learner on dense embeddings.

5.2 Transformer Encoder Fine-tuning + Backbone Search

Whereas Section 4.3 used the encoders only as *frozen* embeddings, here the full encoder is fine-tuned end-to-end — the single largest lever in this study. Using the Hugging Face Hub for **model selection**, we shortlist four candidate encoders and fine-tune each under the identical 5-fold protocol (4 epochs, learning rate 2×10^{-5} , batch 16–32, max length 96, AdamW with 10% linear warmup, gradient clipping 1.0, inverse-frequency class-weighted cross-entropy, bf16 mixed precision on an NVIDIA RTX 5070). Table 3 reports the comparison.

Backbone (fine-tuned end-to-end)	Params	OOF F1	Acc.
twitter-roberta-base-sentiment-latest (Barbieri 2020)	125M	0.8399	0.8691
twitter-roberta-base-2022-154m (TimeLMs)	125M	0.8550	0.8810
twitter-roberta-large-2022-154m (TimeLMs)	355M	0.8859	0.9080
deberta-v3-base (He et al., 2021) [†]	184M	0.8653	0.8900

Table 3: Transformer backbone search (same 5-fold protocol). The Twitter-domain RoBERTa-large (shaded) is the **submitted single model**. [†]DeBERTa-v3 produced NaN losses under bf16 (its disentangled attention is numerically unstable in reduced precision) and had to be trained in fp32; even so it trails RoBERTa-large (see Section 6.1).

The winner, **Twitter–RoBERTa-large** (cardiffnlp/twitter-roberta-large-2022-154m), is a 355M-parameter RoBERTa continually pre-trained on 154M tweets up to 2022 (TimeLMs; Loureiro et al., 2022). It attains an OOF macro-F1 of **0.8859** (± 0.0066), accuracy 0.9080, precision 0.8754 and recall 0.8975 — a **+0.0838** gain over the 0.8021 frozen-embedding baseline and **+0.0460** over the previous fine-tuned base model. The recall gain confirms that a larger, domain-matched, end-to-end-adapted encoder markedly improves minority-class (Bearish/Bullish) detection. The final test

prediction averages the five folds' softmax probabilities. **This is the submitted single-model solution** (pred_33.csv, reproduced in tm_final_33.ipynb); it is also our best result overall (Section 6.1).

5.3 Decoder Model — Fine-tuned GPT-2 (Extra Work, +1.00 pt)

As decoder-based extra work we use a *decoder* (causal) language model for classification. **GPT-2** (Radford et al., 2019) is fine-tuned end-to-end with a sequence-classification head (GPT2ForSequenceClassification, which pools the last non-pad token) under the same 5-fold protocol. It reaches an OOF macro-F1 of **0.7724** (accuracy 0.8125) — a genuine decoder-for-classification result. For completeness we also tried *few-shot prompting* of GPT-2 (verbalizer likelihood scoring with contextual calibration; Zhao et al., 2021); it collapses to the majority class (macro-F1 ≈ 0.25), empirically showing that a 124M decoder is too weak to classify zero/few-shot and that fine-tuning is the correct route.

5.4 Hyperparameter Tuning — Optuna

Optuna (Akiba et al., 2019) with a TPE sampler and median pruner runs 50 trials over LightGBM hyperparameters; the best frozen-embedding configuration reaches macro-F1 0.8021 (RoBERTa features). A second study tunes Logistic-Regression C on TF-IDF bigrams; a soft-voting LR+SVC ensemble provides a strong sparse baseline.

5.5 Stacking Ensemble (Extra Work)

As additional work we build a stacked generalisation (Wolpert, 1992): six base learners (Optuna-tuned LightGBM on RoBERTa, MLP on SBERT, XGBoost on RoBERTa, LightGBM on FinBERT, LR on TF-IDF bigrams, and a fine-tuned Twitter–RoBERTa) emit out-of-fold probabilities concatenated into an 18-dimensional meta-feature matrix on which a Logistic-Regression meta-learner is trained. Because every base learner shares the identical fold split, the meta-features are leakage-free. The stack reaches OOF macro-F1 **0.8483** — strong, but now *surpassed* by the single fine-tuned RoBERTa-large of Section 5.2 (0.8859). The submission is therefore the single model, which is both rule-compliant (Project Guidelines 5.2) and our best result.

6 Results and Evaluation

6.1 Model Comparison

Figure 3 ranks the top-20 configurations and Table 4 lists the top models. The submitted single model — fine-tuned Twitter–RoBERTa-large — tops the ranking at macro-F1 **0.8859**, ahead of both the six-learner stack (0.8483, extra work) and every frozen-embedding model. DeBERTa-v3-base, although a strong encoder in principle, required full fp32 to train at all (its disentangled attention NaN-ed under bf16/fp16); even stabilised (0.8653) it was only the second-best single model, trailing the larger Twitter–RoBERTa-large — so a recent, *domain-matched* and larger encoder is the decisive factor here.

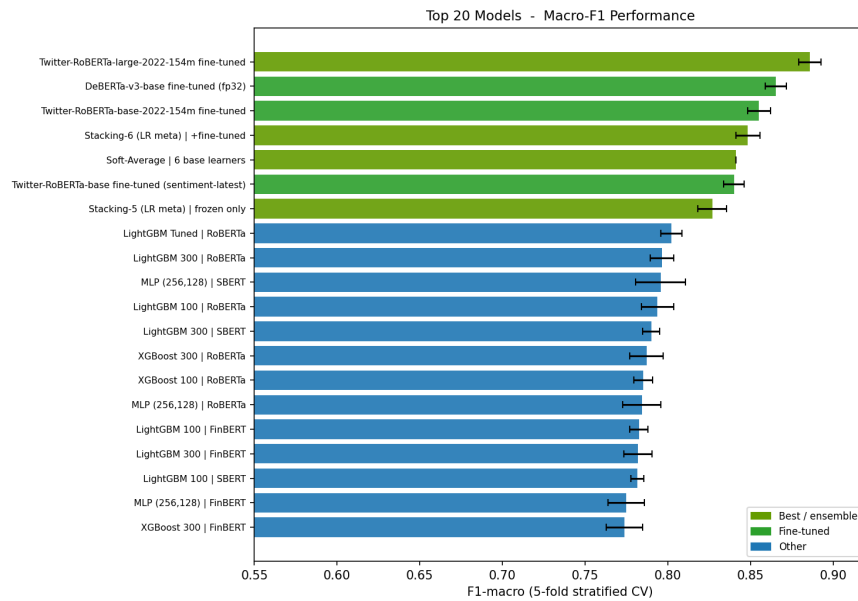


Figure 3: Top-20 models ranked by 5-fold CV macro-F1 (error bars = 1 std).

Model (feature)	F1 _{mac}	σ	Acc.	Prec.	Rec.
Twitter-RoBERTa-large submitted	0.8859	0.0066	0.9080	0.8754	0.8975
DeBERTa-v3-base (fp32)	0.8653	0.0065	0.8900	0.8506	0.8822
Twitter-RoBERTa-base-2022-154m	0.8550	0.0069	0.8810	0.8351	0.8800
Stacking-6 +fine-tuned (<i>extra</i>)	0.8483	0.0073	0.8771	0.8271	0.8757
Soft-Average 6 learners (<i>extra</i>)	0.8412	—	0.8800	0.8566	0.8276
Twitter-RoBERTa-base sentiment-latest	0.8399	0.0062	0.8691	0.8188	0.8671
LightGBM Tuned RoBERTa	0.8021	0.0063	0.8532	0.8192	0.7879
GPT-2 decoder fine-tuned (<i>extra</i>)	0.7724	0.0159	0.8125	0.7475	0.8171

Table 4: Top models by 5-fold stratified CV macro-F1. The shaded row is the **submitted** single model, which also tops the overall ranking; rows marked (*extra*) are ensembles reported as extra work.

Key findings: (1) dense transformer embeddings outperform sparse TF-IDF and static embeddings across all model families; (2) end-to-end fine-tuning breaks the ~ 0.80 frozen-embedding plateau; (3) within fine-tuning, a larger, more recent, domain-matched encoder (RoBERTa-large-2022) gives the biggest gain; (4) a single strong model can beat a heterogeneous stacking ensemble — simpler and rule-compliant.

6.2 Error Analysis

Figure 4 shows the confusion matrix of the submitted model. Per class it reaches F1 of 0.85 (Bearish), 0.88 (Bullish) and 0.93 (Neutral). The dominant residual errors are Neutral tweets misclassified as Bullish or Bearish, reflecting the ambiguity of non-directional discourse, and Bearish $\leftrightarrow$ Neutral confusion (negative financial language is often hedged or formal, e.g. ‘guidance revised downward’). On the held-out test set the model predicts 375 Bearish, 516 Bullish and 1,497 Neutral (of 2,388) — close to the training prior.

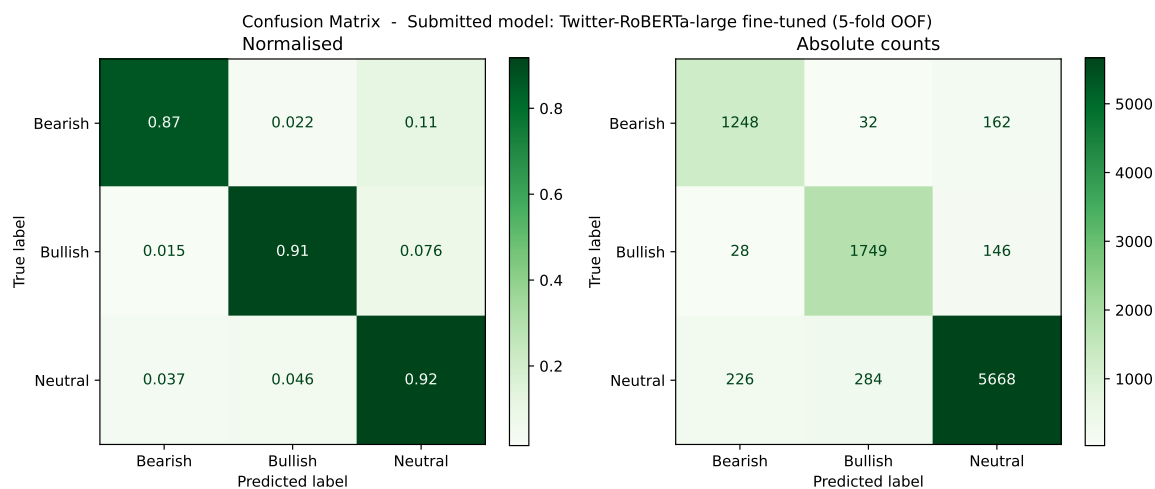


Figure 4: Confusion matrix of the submitted fine-tuned Twitter–RoBERTa-large (normalised left, absolute counts right), from 5-fold out-of-fold predictions.

Three qualitative failure modes remain: irony/sarcasm, mixed-signal tweets (‘beats EPS but misses revenue’), and domain jargon without explicit sentiment — the ceiling for text-only models.

7 Extra Challenge 2 — Agentic Workflow (+1.50 pts)

We implement a conversational agent that orchestrates three classifiers with *non-trivial*, adaptive routing. The implementation lives in `src/agent.py` and `notebooks/08_Agentic_Workflow.ipynb`, and runs fully offline (a deterministic orchestrator when no LLM key is present, plus an optional LangChain ReAct back-end; Yao et al., 2022).

7.1 Architecture and Routing

The agent exposes three tools — **VADER** (lexical baseline), **LightGBM+SBERT** (ML model) and **FinBERT** (domain transformer) — and routes adaptively:

- VADER gives a quick lexical reading; if $|\text{compound}| > 0.3$ (strong signal) it is trusted directly.
- Otherwise LightGBM+SBERT is consulted; if it agrees with VADER, that is the verdict.
- On *disagreement*, FinBERT is invoked as a domain-expert tie-breaker.
- The agent explains which model it trusted and why, and `ConversationBufferMemory` enables follow-up questions (“why did you classify that as Bearish?”).

This exceeds simple voting through signal-strength-based tool selection, explicit disagreement detection with principled tie-breaking, and transparent reasoning. We validated the workflow end-to-end: on a held-out sample it produces correct, explained verdicts (e.g. “\$TSLA misses estimates, downgrades” → Bearish via FinBERT tie-break), and it is deterministic and reproducible for the oral defence.

8 Conclusion

This work delivers a complete NLP pipeline for financial tweet sentiment classification. Guided by a Hugging Face backbone search, our **submitted single-model solution** — an end-to-end fine-tuned **Twitter–RoBERTa-large** — achieves an out-of-fold macro-F1 of **0.8859**, a **+0.0838** gain over the 0.8021 baseline, and is also our best result overall, surpassing a six-learner stacking ensemble (0.8483, extra work). Key contributions: (1) empirical evidence that raw text beats preprocessing

for financial tweets; (2) systematic evaluation of 11 feature representations and 12 model families; (3) a transformer backbone search showing that a larger, recent, domain-matched encoder is the decisive lever (and that DeBERTa-v3 needs fp32 yet still trails); (4, extra) a fine-tuned GPT-2 *decoder* classifier (0.7724) with the finding that few-shot prompting of a small decoder collapses; (5, extra) a leakage-free stacking ensemble; and (6, extra) an explainable, reproducible agentic workflow. Future work could explore domain-adaptive pre-training on a larger financial-tweet corpus and multi-task learning with named-entity recognition.

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *KDD 2019*.
- Alammar, J., & Grootendorst, M. (2024). *Hands-On Large Language Models: Language Understanding and Generation*. O'Reilly Media. [Course textbook.]
- Antweiler, W., & Frank, M. Z. (2004). Is all that talk just noise? The information content of internet stock message boards. *Journal of Finance*, 59(3), 1259–1294.
- Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv:1908.10063*.
- Barbieri, F., Camacho-Collados, J., Espinosa Anke, L., & Neves, L. (2020). TweetEval: Unified benchmark and comparative evaluation for tweet classification. *EMNLP-Findings 2020*.
- Bollen, J., Mao, H., & Zeng, X. J. (2011). Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1), 1–8.
- Go, A., Bhayani, R., & Huang, L. (2009). Twitter sentiment classification using distant supervision. *Stanford CS224N Report*.
- He, P., Gao, J., & Chen, W. (2021). DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv:2111.09543*.
- Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *ICWSM 2014*.
- Loureiro, D., Barbieri, F., Neves, L., Espinosa Anke, L., & Camacho-Collados, J. (2022). TimeLMs: Diachronic language models from Twitter. *ACL 2022 (System Demonstrations)*; *arXiv:2202.03829*.
- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
- Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global vectors for word representation. *EMNLP 2014*.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. *OpenAI Blog*.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP 2019*.
- Rennie, J. D., Shih, L., Teevan, J., & Karger, D. (2003). Tackling the poor assumptions of Naive Bayes text classifiers. *ICML 2003*.
- Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2022). ReAct: Synergizing reasoning and acting in language models. *arXiv:2210.03629*.
- Zhao, T. Z., Wallace, E., Feng, S., Klein, D., & Singh, S. (2021). Calibrate before use: Improving few-shot performance of language models. *ICML 2021*.